

SASIDHAR CHINTAPALLI

Senior Director, Product Architecture | AI & Data Platform Engineering | 17+ Years

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PROFESSIONAL SUMMARY

Engineering leader with 18+ years of experience architecting large-scale distributed data platforms, cloud-native systems, and AI-powered enterprise applications. Proven track record of designing production-grade data layers, custom Spark compute frameworks on Kubernetes, and agentic AI systems that automate the data engineering lifecycle. Adept at leading cross-functional teams from zero-to-one product builds through to enterprise-scale deployment, with deep expertise spanning data engineering, containerized infrastructure, and Generative AI integration.

CORE COMPETENCIES

Data Engineering: Apache Spark, Hadoop, Hive, Snowflake, BigQuery, Databricks, Apache Kafka, PostgreSQL, Kuzu (Graph DB), ETL/ELT Pipelines

AI & LLMs: Agentic AI, GraphRAG, LangGraph, Generative AI, LLM Integration, NL-to-SQL, Hybrid Retrieval (BM25 / pgvector / RRF), LLM Evaluation, ML Pipelines

Cloud & Infra: Kubernetes, Docker, Service Mesh, CI/CD, AWS/Azure/GCP, Cloud-Native Architecture

Languages: Python, Java, TypeScript, SQL, Cypher

Leadership: Product Architecture, System Design, Cross-Functional Team Leadership, 0-to-1 Builds

PROFESSIONAL EXPERIENCE

Uniphore (via acquisition of Infoworks.io)

Senior Director, Product Architecture (Architect) | Jan 2025 – Present | Bengaluru

- Modernizing enterprise data platforms to be **AI-ready** by architecting a 4-layer **data intelligence platform** spanning Data Ingestion, Pre-Processing, AI Catalog (Knowledge Graph + Vector Store + Metadata), Retrieval, and Generation layers—enabling optimized context assembly and augmentation for LLM-powered applications.
- Designed a unified **Data Ingestion Layer** supporting structured sources (SQL/NoSQL databases, APIs, knowledge bases) and unstructured sources (documents, web pages, logs, emails), with a **Pre-Processing pipeline** for entity extraction (NER/LLM), relationship extraction, embedding generation, text chunking, schema mapping, and entity resolution.
- Architected the **AI Catalog Layer** combining a Knowledge Graph (entity-relationship modelling), Vector Store (semantic embeddings), and rich Metadata (source references, timestamps, entity properties, graph references)—forming the backbone for hybrid retrieval and multi-hop reasoning.
- Built a **Hybrid Retrieval pipeline** with graph traversal and subgraph extraction, semantic vector similarity search, hybrid ranking and fusion, and context assembly—delivering optimized, high-fidelity context to the Generation Layer (LLM) for accurate, grounded responses.
- Designed **agentic AI orchestration** with four specialized data agents: Data Engineering Agent (intelligent ETL/ELT automation), Data Discovery Agent (autonomous Catalog and schema discovery), Data Analyst Agent (natural language query interface for reporting/dashboarding requirements)
- Led an **NL-to-SQL accuracy enhancement** program for the Data Analyst Agent: built a hybrid retrieval pipeline fusing BM25 full-text (Postgres), pgvector semantic search and LSH value matching via Reciprocal Rank Fusion + cross-encoder reranking, with Kuzu graph shortest-path traversal over FK relations to resolve join paths.
- Achieved **~72% execution accuracy on the BIRD mini-dev benchmark** (500 questions; 70.9% on the full 1,534-question BIRD dev set) using open-weight models (Gemma 4 31B, Qwen3), via graph-based join-path recovery, compact context assembly, conflict-free layered prompts and a deterministic draft-verify gate, after measuring that multi-agent ensembles and LLM judges did not pay for their cost.
- Built a **reproducible evaluation harness** (official BIRD EX scoring, per-difficulty breakdowns, run manifests with git/config capture, regression/sentinel subsets) and a measured ~1.5-pt noise floor, so every change is accepted on consistent question flips rather than single-run deltas.
- Leading **cross-functional platform engineering** for enterprise-grade security, compliance, and operational excellence at scale, serving both internal services and external customers.
- Maintaining and evolving the **Infoworks data platform**, bridging legacy systems to modern AI-driven workflows with seamless migration paths.

Infoworks.io

Head of Product Architecture, Office of CTO | Jun 2024 – Jan 2025 | Bengaluru

- Core team member driving innovation, design, and development of **Infoworks AI** (ai.infoworks.io), a greenfield SaaS product built from scratch.
- Owned end-to-end product architecture including accuracy, security & compliance, reliability, observability, payments & subscription handling, and cost optimization.
- Guided two engineering teams through full product lifecycle from ideation to release.

Architect | Jan 2021 – May 2024 | Bengaluru

- Architected **Kubernetes-based deployment** of the Infoworks platform with service mesh architecture, containerizing all services for scalable, reliable data pipeline execution.
- Designed and developed **custom Spark compute on Kubernetes**, enabling native ETL job execution with multi-version Spark support.
- Built **SQL workload migration framework** that migrated customers from legacy ETL tools (e.g., Informatica PowerCenter) to Infoworks data pipelines.
- Led migration from **Hadoop-based on-prem deployment to cloud-native architecture**, designing integration support for managed Spark providers (EMR, Dataproc, Databricks) and cloud data warehouses (Snowflake, BigQuery).
- Architected **Data Lakehouse support** and built execution frameworks for complex ETL/ELT workloads, enabling Fortune 500 clients to run thousands of data pipeline jobs at scale.
- Designed a **security-centric control plane / data plane separation** architecture, ensuring enterprise-grade data isolation and governance while scaling workload orchestration across multi-tenant environments.

Principal Software Engineer | Sep 2016 – Jan 2021 | Bengaluru

- Designed and developed a **BigData pipeline engine** with abstraction layers for Hive, Impala, Spark, and SQL data warehouses, replicating full capabilities of traditional ETL tools.
- Designed and developed a **low-code/no-code Transformation Layer** for data lakes following the **medallion architecture** (Bronze → Silver → Gold), enabling business users and data engineers to visually author complex data transformations; deployed across customer data platforms handling thousands of transformation pipelines daily.
- Integrated **AI/ML support for predictive analytics** on structured data within the pipeline framework, enabling inline model training and inference as native pipeline steps for enriched, analytics-ready datasets.

Think & Learn Pvt. Ltd. (Byju's)

Principal Software Engineer | Dec 2015 – Aug 2016 | Bengaluru

- Built **A/B testing framework** from scratch and integrated it with marketing applications.
- Recruited and built an engineering team from the ground up for feature development.

Housing.com

Principal Software Engineer | May 2015 – Nov 2015 | Bengaluru

- Designed and delivered an **e-commerce platform** for real estate (land/plot vertical) from scratch, leading both mobile and web engineering.
- Released two Android apps (Agents and Consumers) and built the hosting platform for all applications.

Zynga

Principal Software Engineer | Aug 2014 – Apr 2015

Senior Software Engineer | May 2011 – Aug 2014 | Bengaluru

- Led feature development across multiple titles (CastleVille, Empires & Allies, DrawSomething, Treasure Isle) as Pod Lead, driving client and server-side engineering.
- Delivered **ad monetization** features and revenue growth; drove performance optimizations across client and server stacks using HiDef, Memcache, and Membase.
- Built a **Content Management System (CMS)** for automated game content delivery.

Yahoo

Senior System Engineer | Jun 2008 – May 2011 | Bengaluru

- Developed **Site Builder Web Service** for microsite creation with module-level validation and preview.
- Built SOA-based **publishing platform** for site versioning, branch-level content serving, and template-driven page generation.
- Worked on Global Travel backend feed processing platform and release engineering.

OPEN SOURCE

Kuzu Explorer | VS Code extension for the Kuzu graph database | TypeScript, Node.js | github.com/gaintlabs/kuzu-viewer

- Built and open-sourced (MIT) a **VS Code extension** to browse Kuzu graph databases in the editor: schema tree of node/rel tables, sortable table-data grid, Cypher query panel and an interactive Cytoscape graph view; ported from an original FastAPI web app.
- Isolated the native Kuzu addon in a **system-Node worker process over newline-delimited JSON-RPC** to avoid Electron ABI mismatches; read-only by default with a write-query guard and transparent snapshot fallback when the database is locked.
- Set up a **GitHub Actions release pipeline** building platform-specific VSIX packages (Linux x64, Windows x64, macOS x64/arm64) with an automated VS Code Marketplace publish step.

EDUCATION

M.Tech, Computer Science & Engineering | International Institute of Information Technology | 2006 – 2008

B.Tech, Computer Science | Sasi Institute of Technology & Engineering | 2002 – 2006

CERTIFICATIONS

- Generative AI with Large Language Models
- Machine Learning Specialization
- Red Hat Certified System Administrator (RHCSA)
- MySQL 5.6 Database Administrator